Algorithms Design and Analysis - HW I

Ronen Finkelstein

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Due to: 20/11/2025

Please submit the assignment in pairs, and make sure to include both of your names and ids on the PDF. 5 bonus points are given for a typed out solution, and my respect to whoever writes in LATEX. Make sure to prove each claim you use, unless it was seen in class. You may contact me for every question you have about the H.W.

Problem 1:

Let $f(n)$ be defined by:

$$f(n) = \begin{cases} n + f(\alpha n) + f(\beta n), & \text{if } n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Where $0 < \alpha, \beta < 1$. Give a tight asymptotic for $f(n)$ in each case:

1. $\alpha + \beta < 1$
2. $\alpha + \beta = 1$
3. $\alpha + \beta > 1$

Problem 2:

1. Prove for any $1 < \alpha \in \mathbb{N}$: $log^a n = o(\sqrt{n})$

2. Next there are 5 functions, order them in asymptotic ascending order. I.e.

$$f_1, f_2, f_3, f_4, f_5 : f_1 = O(f_2) = O(f_3) = O(f_4) = O(f_5)$$

The functions are:

$$log(n^n), (logn)^n, log(n^{6006}), (logn)^{6006}, loglog(6006n)$$

Problem 3:

A matrix M with r rows and c columns containing **distinct** integers is said to be *sorted* if each row and column is sorted. Namely, for every $1 \geq i < r, 1 \geq j < c$ it holds that

$$M(i,j) < M(i,j+1) \text{ and } M(i,j) < M(i+1,j)$$

For example, the following matrix M is a sorted matrix:

$$M = \begin{bmatrix} 5 & 8 & 11 & 23 \\ 6 & 9 & 14 & 25 \\ 10 & 15 & 18 & 31 \\ 30 & 32 & 40 & 50 \end{bmatrix}$$

1. Given a sorted matrix, which cell contains the minimum element? Prove it. Namely, if you claim that the minimum element is in cell $M(a,b)$ you need to show that for any $(a',b') \neq (a,b) \implies M(a,b) < M(a',b')$

2. Given a sorted matrix M with n rows and n columns and an integer z, we want to check if z is in M. A student suggested the following algorithm:

   (a) Compare z with an element w in the middle of the matrix. Namely, $w = M(x_{mid}, y_{mid})$ where $x_{mid} = \lfloor \frac{n+1}{2} \rfloor$, $y_{mid} = \lfloor \frac{n+1}{2} \rfloor$

   (b) Divide M to four sub-matrices A, B, C, D of size (roughly) $n/2n/2$:

   $$A = M(1, ..., x_{mid})(1, ..., y_{mid})$$
   $$B = M(1, ..., x_{mid})(y_{mid}+1, ..., n)$$
   $$C = M(x_{mid}+1, ...n)(1, ..., y_{mid})$$
   $$D = M(x_{mid}+1, ..., n)(y_{mid}+1, ..., n)$$

   (c) If $z = w$, return **Found**.

   (d) If $z < w$, we look for z recursively in the following sub-matrices (fill in the blank): _____

   (e) If $z > w$, we look for z recursively in the following sub-matrices (fill in the blank): _____

3. Write a pseudo-code implementing the above algorithm and analyze its running time. Namely, let $T(n)$ denote the worst-case running time of the algorithm on a matrix of size nn. Write the recurrence relation and solve it.

4. Write an algorithm checks if z is an element of a sorted matrix M. Your algorithm should have runtime which is better than $O(n^{1.001})$

Problem 4:

Given an array $arr[]$ of rope lengths, connect all ropes into a single rope with the minimum total cost. The cost to connect two ropes is the sum of their lengths. Example: Input $arr[] = [4, 3, 2, 6]$
Output: 29 Explanation: Minimum cost to connect all ropes into a single rope is
Connect ropes 2 and 3 - $[4, 5, 6]$, cost = 5
Connect ropes 4 and 5 - $[9, 6]$, cost = 9
Connect ropes 9 and 6 - $[15]$, cost = 15
Total cost = 5 + 9 + 15 = 29

Problem 5:

Given a list $D = d1, d2, ..., dn$ of n natural numbers, Design a Greedy algorithm that decides wether there exists an undirected graph $G = (v1, v2, ..., vn, E) - s.t. \forall i : degree(vi) = di$. G should be a simple graph which means that it should not contain self loops or multiple edges between the same pair of nodes. if exists such a graph the algorithm should return it, otherwise it returns "FALSE". Justify correctness of your solution and analyze running time. Example A: $D = (1, 1, 1, 1)$. The graph with the set of edges $(v1, v2), (v3, v4)$ may be returned by the algorithm. Example B: $D = (4, 3, 1, 1)$. There is no graph with four having theses degrees, then the algorithm in this case returns "FALSE".